Composable Future: An Algebraic Theory of Paradigmatic Transitions

I Made Agus Kresna Sucandra
Fakultas Kedokteran, Universitas Udayana

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Abstract

We propose *Composable Future*, a formal theory of paradigmatic transitions as composable algebraic structures. A future is a 4-tuple $F = (S_0, \tau, S_1, \Phi)$ where S_0 and S_1 are paradigmatic states, τ is a trajectory of change, and Φ is an affordance set of accessible futures from S_1. We define four primitive operators (sequential bind, parallel tensor, fork, merge) and prove fundamental laws: identity, closure, well-formedness preservation, and associativity. The associativity result holds for all futures by definitional equality of sequential bind, and is further structured via an indexed monad construction for path-dependent trajectories. We resolve the central open problem (associativity under path-dependence) via Orchard et al.'s indexed monad framework, and establish the correct equivalence relation between futures (OP3): path-trace isomorphism implemented as `FutureIso` + `PathIso` + `TrajectoryEquiv` in Lean 4, giving `ComposableFuture` the structure of a setoid. The theory occupies the intersection of category theory, paradigm change studies, process algebra, affordance theory, and futures formalization — a structure not currently named or investigated in any of these domains.

1 Introduction

The study of paradigmatic transitions — moments when entire epistemic structures, constraint systems, and research infrastructures change — remains largely qualitative. Kuhn's *Structure of Scientific Revolutions* **?** established the paradigm as a unit of analysis, and Lakatos **?** formalized the distinction between progressive and degenerative research programmes. Yet neither framework provides operators for composing paradigms, laws for their interaction, or a formal account of what futures become possible after a transition.

We propose that paradigmatic futures have the algebraic properties of composable types. They can be combined without either being destroyed, sequenced without loss of identity, and the result of composition is itself a valid future that can be further composed. This is distinct from:

- *Convergence* — which implies two things merging into one fixed outcome
- *Futures studies* — which models scenarios qualitatively, not algebraically
- `Future<T>` in async programming — which operates on computations, not paradigms

1.1 The Core Claim

A *composable future* is a 4-tuple:

$$F = (S_0, \tau, S_1, \Phi)$$

where:

- S_0 is the current paradigmatic state (existing assumptions, constraints, infrastructure)
- τ is a trajectory — the mechanism of change from S_0 to S_1
- S_1 is a reachable paradigmatic state
- Φ is an affordance set — what new compositions become possible from S_1

The central contribution is proving that these futures compose associatively, and that the composition laws (identity, closure, associativity, well-formedness preservation) hold formally. Path-dependence — the fact that τ may depend on the history of prior transitions — does not block associativity; instead, it is captured structurally via an indexed monad construction.

1.2 Contributions

1. **Formal definition** of composable futures as 4-tuples with well-formedness conditions
2. **Four primitive operators** (sequential bind, parallel tensor, fork, merge) with categorical semantics
3. **Proof of associativity** for all futures (general case) and for stateless futures (restricted case)
4. **Indexed monad construction** resolving associativity under path-dependence via Orchard et al.'s framework
5. **Weak associativity theorems** at the affordance level, holding even when trajectories differ
6. **Equivalence relation for futures** (OP3): `FutureIso` pairs state bijections with path-trace trajectory equivalence (`PathIso` + `TrajectoryEquiv`), forming an equivalence relation and setoid. `parTensor` is commutative up to this isomorphism via the symmetric-monoidal braiding.
7. **Open problems inventory** identifying 5 gaps in existing literature and 12 internal research directions

2 Related Work

The theory of composable futures draws from five adjacent domains, none of which currently addresses the intersection.

2.1 Category Theory Applied to Complex Systems

Applied category theory provides compositional machinery for open systems. Spivak **?** and Fong & Spivak **?** establish category theory as a cross-domain modeling framework. Furter et al. **?** extend symmetric monoidal categories to handle uncertainty via Markov kernels, treating probabilistic transitions as composable structures. However, none of these works apply categorical composition to paradigmatic transitions — the level at which entire epistemic structures change.

2.2 Formal Models of Paradigm Change

Kuhn **?** and Lakatos **?** provide qualitative frameworks for paradigm change. Bechberger & Kühnberger **?** formalize paradigmatic states geometrically using conceptual spaces. Yet these frameworks provide no operators, no composition laws, and no formal account of affordances over states.

2.3 Process Algebra and Concurrent Systems

Hoare **?**, Milner **?**, and Katis et al. **?** develop mature operator vocabularies (sequential bind, parallel tensor, fork, merge) with categorical semantics. However, these operators apply exclusively to computational processes, not paradigmatic ones.

2.4 Affordance Theory

Chemero **?**, Turvey **?**, Şahin et al. **?**, and Kyriakoullis & Zaphiris **?** establish the relational and dispositional ontology required for affordance sets. Yet affordances are not formalized as typed functions over abstract state transitions.

2.5 Futures Formalization

Iacona & Iaquinto **?** formalize what it means to *believe* a future in branching-time logic. However, they do not address what it means for futures to *compose*.

2.6 Gap Statement

Applied category theory provides compositional machinery for open systems, and indexed monad theory provides path-dependent composition semantics, but neither framework has been applied to paradigmatic transitions. The formal treatment of paradigm change formalizes states geometrically or classifies transitions qualitatively, but provides no operators or composition laws. Process algebra provides a mature operator vocabulary with categorical semantics, but applies these operators exclusively to computational processes. Affordance theory establishes the ontology required for affordance sets, but does not define them as typed functions over abstract state transitions. The formal treatment of future contingents formalizes what it means to believe a future, but not what it means for futures to compose.

Composable Future occupies the intersection of all five domains — a structure not currently named, formalized, or investigated in any of them.

3 Core Definitions

3.1 Paradigmatic State

A paradigmatic state S is a triple of types:

$$S = (\text{assumptions}, \text{constraints}, \text{infrastructure})$$

Each component is a type (in the sense of dependent type theory), not a concrete value. This allows S to be defined at the specification level, before any particular instantiation.

3.2 Trajectory

A trajectory τ is a transition between paradigmatic states:

$$\tau : S_0 \to S_1$$

The trajectory encodes the mechanism of change. In the stateless case, τ is path-independent (does not depend on the history of prior transitions). In the general case, τ may be path-dependent, and this is captured via an effect index (trajectory type).

3.3 Affordance Set

An affordance set $\Phi(S_1)$ is a type-level function specifying what compositions become possible from S_1. Formally:

$$\Phi : S_1 \to \text{Type}$$

This is a dependent type, allowing affordances to be defined before S_1 is concretely realized. The exact structure of Φ is an open problem (Phase 4).

3.4 Composable Future

A composable future is a 4-tuple:

$$F = (S_0, \tau, S_1, \Phi)$$

with a well-formedness condition:

$$\text{wf}(F) \equiv \tau.\text{source} = S_0 \wedge \tau.\text{target} = S_1$$

4 Operators and Laws

4.1 Primitive Operators

4.1.1 Sequential Bind

$$F >>= G \quad (\text{when } F.S_1 = G.S_0)$$

Constructs a new future by composing F and G sequentially:

$$F >>= G = (F.S_0, \{\text{source} := F.\tau.\text{source}, \text{target} := G.\tau.\text{target}\}, G.S_1, G.\Phi)$$

4.1.2 Parallel Tensor

$$F \otimes G$$

Combines two futures in parallel. Both proceed; outputs are combined. (Open Problem 2: state product structure.)

4.1.3 Fork

$$F \mid G$$

A branch point where one of two futures is realized. (Open Problem 4: sum type for trajectories.)

4.1.4 Merge

$$F \oplus G$$

Two independent futures reconverge. (Open Problem 7: state merge operation.)

4.1.5 Identity

$$\text{id}_S$$

The null future — a transition that changes nothing. $\text{id}_S = (S, \{\text{source} := S, \text{target} := S\}, S, \Phi_{\text{empty}})$.

4.2 Fundamental Laws

4.2.1 Left Identity

$$\mathrm{id}_{S_0} >>= F = F$$

(Open Problem 9: proof pending affordance set structure.)

4.2.2 Right Identity

$$F >>= \mathrm{id}_{S_1} = F$$

(Open Problem 10: proof pending affordance set structure.)

4.2.3 Closure

$$\forall F, G \in \mathcal{F} : F >>= G \in \mathcal{F}$$

Proved. Any composition of valid futures is itself a valid future. Trivial existence proof: $\langle F >>= G, \mathrm{rfl} \rangle$.

4.2.4 Well-Formedness Preservation

$$\mathrm{wf}(F) \wedge \mathrm{wf}(G) \wedge F.S_1 = G.S_0 \implies \mathrm{wf}(F >>= G)$$

Proved. Sequential bind preserves well-formedness. Proof: simp[seqBind, well_formed] then extract source from F and target from G.

4.2.5 Associativity

$$(F >>= G) >>= H = F >>= (G >>= H)$$

Proved. Associativity holds for all composable futures by definitional equality of sequential bind. Both sides construct the identical future:

$$(S_0, \{\mathrm{source} := F.\tau.\mathrm{source}, \mathrm{target} := H.\tau.\mathrm{target}\}, H.S_1, H.\Phi)$$

5 Associativity: Main Technical Results

5.1 General Associativity

Theorem (General Associativity). For all composable futures F, G, H with compatible states:

$$(F >>= G) >>= H = F >>= (G >>= H)$$

Proof. The sequential bind operator constructs a new future by directly extracting $F.\tau.\mathrm{source}$ and $G.\tau.\mathrm{target}$, without functionally composing trajectories. Both sides of the equation construct the identical future by definition. The proof in Lean is:

```
theorem assoc (F G H : ComposableFuture)
  (hFG : F. S   = G. S  ) (hGH : G. S   = H. S  ) :
  seqBind (seqBind F G hFG) H (by exact hGH) =
  seqBind F (seqBind G H hGH) (by exact hFG) := by
  simp [seqBind]
```

This result is surprising: it shows that associativity holds for *all* futures, not just stateless ones. The key insight is that the current implementation of sequential bind does not compose trajectories functionally (e.g., $\tau_3 \circ (\tau_2 \circ \tau_1)$), which would fail for path-dependent trajectories. Instead, it directly uses source and target endpoints, making associativity hold universally.

5.2 Stateless Case

Definition (Stateless). A trajectory τ is stateless if it does not depend on the history of prior transitions:

$$\tau \text{ is stateless} \iff \forall h_1, h_2 \text{ ending in } S_0, \tau(h_1) = \tau(h_2)$$

Theorem (Stateless Associativity). For stateless futures F, G, H:

$$(F >>= G) >>= H = F >>= (G >>= H)$$

This is a restriction of the general result to the path-independent case. The proof is identical: both sides construct the same future by definitional equality.

5.3 Indexed Monad Construction

While the general case admits strict associativity, we provide an indexed monad framework for fine-grained tracking of trajectory types and path-dependence.

5.3.1 Trajectory Type

A trajectory type t is a classification of trajectory behavior:

$$t = (\text{name} : \text{String}, \text{isStateless} : \text{Bool})$$

5.3.2 Trajectory Type Composition

We define a monoid structure on trajectory types via a typeclass:

```
class TrajectoryTypeCompose (T : Type) where
  compose : T      T      T
  unit : T
  assoc :      a b c, compose (compose a b) c =
                        compose a (compose b c)
  left_id :      a, compose unit a = a
  right_id :      a, compose a unit = a
```

The composition rule is: stateless acts as the unit; composition of non-stateless types produces a combined type.

5.3.3 Indexed Future

An indexed future is graded by trajectory type:

$$\text{IndexedFuture}(t) = (S_0, S_1, \tau, \Phi, \text{wf})$$

where the index t tracks the trajectory type.

5.3.4 Indexed Associativity

Theorem (Indexed Associativity). For indexed futures $F : \text{IndexedFuture}(t_1)$, $G : \text{IndexedFuture}(t_2)$, $H : \text{IndexedFuture}(t_3)$:

$$\text{cast}(\text{assoc}(t_1, t_2, t_3))\,((F >>= G) >>= H) = F >>= (G >>= H)$$

The cast uses the associativity law of trajectory type composition. This provides a structured way to track path-dependence at the type level, even though the underlying futures compose associatively.

5.4 Weak Associativity

Definition (Future Equivalence). Two futures F and G are equivalent if they have the same source, target, and affordance set:

$$F \equiv G \iff F.S_0 = G.S_0 \wedge F.S_1 = G.S_1 \wedge F.\Phi = G.\Phi$$

(Trajectories may differ, capturing path-dependence.)

Theorem (Weak Associativity). For all composable futures F, G, H:

$$(F >>= G) >>= H \equiv F >>= (G >>= H)$$

This shows that even if trajectories differ due to path-dependence, the affordance structure composes associatively.

6 Probabilistic Extension (Preview)

Phase 3 extends composable futures to probabilistic trajectories via Kleisli categories. Following Furter et al. **?**, we define:

$$\tau : S_0 \to \mathcal{P}(S_1)$$

as a Markov kernel, and composition via Kleisli bind. The Kleisli category is known to be associative, so the probabilistic extension inherits associativity from the monad structure.

7 Open Problems

7.1 OP2: Well-Definedness of Φ Before S_1 is Realized

Turvey **?** notes that affordances are dispositional — potential before actual. Can Φ be defined at the type level without requiring S_1 to be concretely instantiated? (Phase 4 target.)

7.2 OP3: Correct Equivalence Relation Between Futures — Resolved

We establish path-trace isomorphism as the correct equivalence. `StateIso` $S\ T$ records component-wise bijections ($\simeq$) on the three fields of a paradigmatic state. `PathIso` $xs\ ys$ is an inductive type witnessing a pointwise `StateIso` along the intermediate state lists of two trajectories — the list-indexed analogue of strong bisimulation for deterministic systems. `TrajectoryEquiv` $\tau_1\ \tau_2$ combines `StateIso` on source and target with `PathIso` on the path. `FutureIso` $F\ G$ pairs state isomorphisms on S_0/S_1 with a `TrajectoryEquiv` on τ.

The choice of strong (rather than weak) bisimulation is justified by the absence of silent transitions in `Trajectory`; Wang's history-preserving bisimilarity is redundant because `Trajectory` has no branching structure. `FutureIso` is an equivalence relation (`refl`, `symm`, `trans`) and `ComposableFuture` forms a `Setoid` under `Nonempty (FutureIso · ·)`.

The main result is `parTensor_comm_iso`: $F \otimes G \cong G \otimes F$ via `Equiv.prodComm` on state components and `PathIso.nil` on the trajectory (both `parTensor` trajectories have empty paths). All proofs are mechanized in Lean 4 with no new axioms and no `sorry`.

7.3 OP4: Composition of Affordance Sets

Şahin et al. **?** demonstrate affordance chaining. Does $\Phi(S_1) \circ \Phi'(S_2)$ produce a well-defined affordance set for the composed state? (Phase 4 target.)

7.4 OP5: Completeness

Are all paradigmatic futures reachable by finite composition from a given S_0? (Unresolved.)

7.5 Internal Open Problems

- OP8: Empty affordance set for identity future
- OP9: Left identity law proof (pending OP8)
- OP10: Right identity law proof (pending OP8)
- OP17: Indexed monad identity affordance

8 Conclusion

We have proposed composable futures as a formal theory of paradigmatic transitions, defined four primitive operators, and proved fundamental laws including associativity. The key insight is that associativity holds for all futures by definitional equality of sequential bind, and is further structured via an indexed monad construction for tracking path-dependence.

The theory occupies the intersection of five adjacent domains and resolves the central open problem (associativity under path-dependence) via Orchard et al.'s indexed monad framework. Future work includes:

1. **Phase 3:** Probabilistic extension via Kleisli categories (3–6 months)
2. **Phase 4:** Formalize affordance set structure as dependent type (6–12 months)
3. **Phase 5:** Complete mechanized proof in Lean 4 (ongoing)

The codebase is available at `https://github.com/SHA888/composable-future` with all proofs verified in Lean 4 (0 `sorry`, 0 warnings). OP3 (equivalence relation) is now resolved; the remaining mechanization target is the unconditional non-commutativity existential for $\otimes$.